

Gen-SHM: Physics-Informed Generative Surrogate for Drone Wing Structural Integrity

Track 1: Scientific ML + Generative AI

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Abstract

This work propose a physics-informed generative surrogate to create synthetic vibration data for drone wing damage detection. We have embedded the Euler-Bernoulli beam equation into a parametric neural network. The model generates physically consistent acceleration time histories for arbitrary damage scenarios. This approach addresses the critical lack of failure data for training structural health monitoring systems and enables zero-shot damage identification without destructive testing.

Problem Statement

The development of robust damage detection algorithms for drone wings requires large datasets of vibration responses under diverse damage conditions, such as micro-cracks and delamination. However, obtaining such data experimentally is impractical due to the cost, time, and risk involved in inducing structural failures. Conventional simulation approaches, such as Finite Element Analysis (FEA), are computationally expensive and unsuitable for real-time or edge deployment. This creates a critical gap between the data requirements of modern AI-based structural health monitoring systems and the limited availability of physically realistic damage data.

Key challenge: To develop a **physics-informed generative surrogate model** capable of rapidly producing physically consistent sensor data for arbitrary damage levels. This enables the training of onboard structural health monitoring systems without requiring destructive testing or extensive real-world failure data.

Why This Problem Matters ?

- **Safety critical:** Invisible fatigue cracks in composite wings are a leading cause of mid-air failure in delivery drones.
- **Data scarcity:** Real-world structural failure data is extremely limited. Standard generative models such as GANs and diffusion models can produce realistic looking signals but often violate fundamental physical laws, such as conservation of energy and momen-

tum. As a result, their outputs may be physically inconsistent and unreliable for engineering analysis and safety critical decision making.

- **Impact:** A physics-grounded generative model unlocks *zero-shot* damage detection. Therefore, drones can recognize failures they have never experienced in flight through stiffness variations in their composite beams.

Dataset & Assumed Data Source

The model relies on two sources:

1. **Physics foundation:** The **Euler–Bernoulli beam theory** is a fundamental model in solid mechanics is used as the governing equation to represent the vibration behavior of the wing structure.
2. **Calibration data:** Sparse accelerometer readings from a healthy drone (e.g., 3–5 sensors along the wing) to identify baseline material properties and validate the healthy solution.

From these, the generative model produces a synthetic dataset of 100,000+ acceleration time-histories corresponding to varying stiffness reductions d (damage location and severity).

Proposed Modeling Approach

We design a **parametric physics-informed neural network (PINN)** that acts as a generative engine. Unlike a standard solver, it learns the *solution operator* across a continuous space of damage parameters.

A. Governing Equation

The wing is modeled as a cantilever beam with spatially varying stiffness. The network must satisfy the dynamic Euler-Bernoulli equation:

$$\rho A \frac{\partial^2 u}{\partial t^2} + c \frac{\partial u}{\partial t} + \frac{\partial^2}{\partial x^2} \left(EI(x; d) \frac{\partial^2 u}{\partial x^2} \right) = 0, \quad (1)$$

where $u(x, t)$ is the vertical displacement, ρA the mass per unit length, c the damping coefficient, and $EI(x; d)$

the stiffness field parameterized by a damage variable d . Damage is encoded as a local reduction in Young’s modulus: $EI(x; d) = EI_0(1 - d \cdot \phi(x))$ with $\phi(x)$ a spatial influence function.

B. Generative Architecture

The neural network $\mathcal{G}$ takes as inputs the space-time coordinates (x, t) and a sampled damage parameter d (drawn from a uniform distribution over plausible crack locations and severities). It outputs the predicted vibration response:

$$\hat{u}(x, t) = \mathcal{G}(x, t; d). \quad (2)$$

For each d , the network can generate an arbitrary number of independent vibration signals by varying the implicit random seed (e.g., through dropout or by sampling initial condition perturbations).

C. Physics-Informed Loss

Training minimizes a composite loss that combines data fidelity and physics compliance:

1. Data loss (calibration) ensures the model matches the healthy drone baseline ($d = 0$):

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N |\hat{u}(x_i, t_i; 0) - u_{\text{obs}}(x_i, t_i)|^2.$$

2. Physics residual penalizes violations of Eq. (1) for any damage parameter:

$$\mathcal{L}_{\text{physics}} = \frac{1}{M} \sum_{j=1}^M \|\mathcal{R}[\hat{u}](x_j, t_j; d_j)\|^2,$$

where $\mathcal{R}$ is the left-hand side of Eq. (1) evaluated via automatic differentiation. The total loss is $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{physics}}$.

D. Generative Capability

Now the model acts as a surrogate generator for the desired damage parameter d . It can produce countless vibration samples by feeding different (x, t) queries. These samples respect Newton’s laws because the physics residual is minimized during training.

Scientific Assumptions

1. Linear elasticity: The wing material is modeled as linearly elastic obeying Hooke’s law which assumes a proportional relationship between stress and strain up to failure.

2. Euler-Bernoulli validity: Since the wing’s length-to-thickness ratio exceeds 10:1, shear deformation and rotational inertia effects are negligible. Thus, Timoshenko effects are ignored, and the Euler-Bernoulli beam model is used to ensure computational efficiency without significant loss of accuracy.

3. Damage as stiffness reduction: Structural damage is modeled purely as a local reduction in Young’s modulus E instead of explicit geometric discontinuities (mesh breakage). This approach is widely used in structural health monitoring to model early-stage cracks, where stiffness degradation precedes complete structural failure.

4. Damping model: Viscous damping ($c \partial u / \partial t$) is modeled as constant and is estimated from healthy-state calibration data, assuming damping properties remain unchanged under damage.

Expected Outcome

The result is a **physics-verified generative tool** that outputs clean, labeled vibration datasets for structural damage analysis.

- **Input:** Generate 100 samples of a wing with a 20% crack at the root.
- **Output:** 100 distinct physically consistent acceleration time-history signals at the sensor locations.

The generated synthetic datasets are used to train a lightweight edge classifier (e.g., a 1D CNN), enabling accurate damage detection even for previously unseen damage scenarios. By embedding governing physics into the generative process, the surrogate overcomes real-data scarcity and enables robust, generalizable structural health monitoring (SHM) for autonomous aerial systems.

Project Demo

A demonstration video showcasing the physics-informed generative surrogate, synthetic vibration signal generation, and damage scenario visualization is available at: **Video**